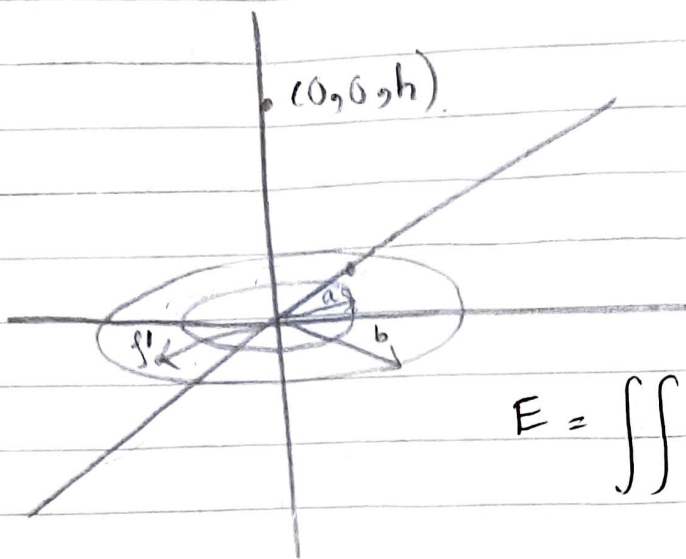


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EMT - MIDTERM



$$\mathbf{r} = h \mathbf{a}_z$$

$$\bar{\mathbf{r}} = f' \mathbf{a}_f$$

$$E = \iint \frac{\rho_s (\mathbf{r} - \bar{\mathbf{r}})}{4\epsilon_0 \pi |\mathbf{r} - \bar{\mathbf{r}}|^3} ds$$

$$\mathbf{r} - \bar{\mathbf{r}} = h \mathbf{a}_z - f' \mathbf{a}_f$$

$$|\mathbf{r} - \bar{\mathbf{r}}| = \sqrt{h^2 + f'^2}$$

$$ds = f' d\phi df'$$

$$E = \iint \frac{\rho_s (h \mathbf{a}_z - f' \mathbf{a}_f)}{4\pi \epsilon_0 (\sqrt{h^2 + f'^2})^3} \cdot f' d\phi df'$$

$\mathbf{a}_f$ component of E ignored as it be cancelled out due to symmetry

$$E = \frac{\rho_s h \mathbf{a}_z}{4\pi \epsilon_0} \iint \frac{f'^2}{(\sqrt{h^2 + f'^2})^3} d\phi df'$$

$$E = \frac{\rho_s h \mathbf{a}_z}{4\pi \epsilon_0} \int_{\phi=0}^{2\pi} d\phi \int \frac{f'^2}{(\sqrt{h^2 + f'^2})^3} df'$$

$$\text{let } f' = h \tan \theta \quad df' = h \sec^2 \theta \cdot d\theta$$

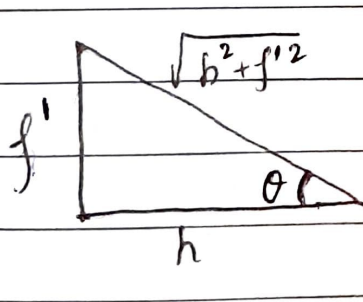
$$\int_a^b \frac{f^2}{(\sqrt{h^2 + f'^2})^3} df = \int_{\theta_1}^{\theta_2} \frac{h^2 \tan^2 \theta h \sec^2 \theta d\theta}{(h^2 + h^2 \tan^2 \theta)^3}$$

$$\int_{\theta_1}^{\theta_2} \frac{h^3 \tan^2 \theta \sec^2 \theta d\theta}{h^3 \sec^5 \theta} = \int_{\theta_1}^{\theta_2} \frac{\sec^2 \theta - 1}{\sec \theta} d\theta$$

$$\int_{\theta_1}^{\theta_2} (\sec \theta - \cos \theta) d\theta$$

$$\int \sec \theta d\theta \text{ as seen from google} = \ln |\sec \theta + \tan \theta|$$

$$\ln |\sec \theta + \tan \theta| - \sin \theta \Big|_{\theta_1}^{\theta_2}$$



$$\sec \theta = \frac{1}{\cos \theta} = \frac{\sqrt{h^2 + f'^2}}{h} \quad \tan \theta = \frac{f'}{h}$$

$$\sin \theta = \frac{f'}{\sqrt{h^2 + f'^2}}$$

$$\left[\ln \left| \frac{\sqrt{h^2 + f'^2} + f'}{h} \right| - \frac{f'}{\sqrt{h^2 + f'^2}} \right]_a^b$$

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$$\left[\ln \left| \frac{\sqrt{h^2+b^2}+b}{h} \right| - \frac{b}{\sqrt{h^2+b^2}} - \ln \left| \frac{\sqrt{h^2+a^2}+a}{h} \right| + \frac{a}{\sqrt{h^2+a^2}} \right]$$

$$\ln \left| \frac{\sqrt{h^2+b^2}+b}{\sqrt{h^2+a^2}+a} \right| - \frac{b}{\sqrt{h^2+b^2}} + \frac{a}{\sqrt{h^2+a^2}}$$

$$E = \frac{Ch}{2\epsilon_0} \left[\ln \left| \frac{\sqrt{h^2+b^2}+b}{\sqrt{h^2+a^2}+a} \right| - \frac{b}{\sqrt{h^2+b^2}} + \frac{a}{\sqrt{h^2+a^2}} \right] \hat{z}$$

Q2. $f_v = f_0 \left(\frac{a^3 - r^3}{a^3} \right)$ $f_0 = \text{constant}$

a) total charge $Q = \int_V f_v dv$

$$\int_{r=0}^a \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} f_0 \left(\frac{a^3 - r^3}{a^3} \right) r^2 \sin \theta dr d\theta d\phi$$

$$\frac{f_0}{a^3} \int_{r=0}^a (a^3 - r^3) r^2 dr \int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin \theta d\theta$$

$$\frac{4\pi f_0}{a^3} \int_{r=0}^a (a^3 r^2 - r^5) dr$$

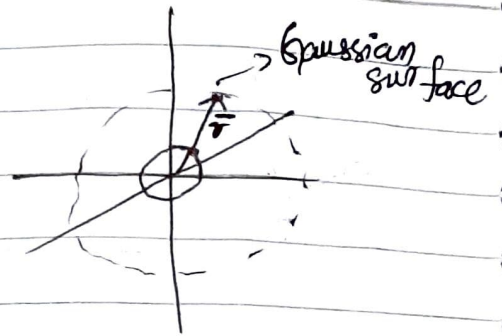
$$\frac{4\pi f_0}{a^3} \left[\frac{a^3 r^3}{3} - \frac{r^6}{6} \right]_0^a$$

$$\frac{4\pi f_0}{a^3} \left[\frac{a^6}{3} - \frac{a^6}{6} \right] = \frac{4\pi f_0}{a^3} \times \frac{a^6}{6} = \frac{2\pi f_0 a^3}{3} C$$

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b). E outside the nucleus.

Applying Gaussian of radius $r > a$



By Gauss's law.

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_V \rho \, dv = \text{charge enclosed.}$$

$$\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \epsilon_0 E_r \cdot ds = \frac{2}{3} \pi \int_0^a a^3$$

$$\int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \epsilon_0 E_r \times 2\pi \cdot r^2 \sin\theta \, d\theta \, d\phi \, 2\pi = \frac{2}{3} \pi \int_0^a a^3$$

$$\epsilon_0 E_r r^2 \underbrace{\int_{\phi=0}^{2\pi} d\phi \int_{\theta=0}^{\pi} \sin\theta \, d\theta}_{4\pi} = \frac{2}{3} \pi \int_0^a a^3$$

E_r

$$E_r = \frac{\frac{2}{3} \pi \int_0^a a^3}{\epsilon_0 r^2 4\pi} = \frac{\int_0^a a^3}{6 \epsilon_0 r^2}$$

$$E = E_r \cdot 2\pi = \frac{\int_0^a a^3}{6 \epsilon_0 r^2} \cdot 2\pi$$

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$$V = - \int_L E \cdot dl \quad dl = dr \, 2\pi r$$

$$V = - \int_L \frac{\int_0^a a^3}{6\epsilon_0 r^2} \, dr \, 2\pi r$$

$$V = - \int_{\infty}^r \frac{\int_0^a a^3}{6\epsilon_0 r'^2} \, dr' = - \frac{\int_0^a a^3}{6\epsilon_0} \int_{\infty}^r \frac{1}{r'^2} \, dr'$$

$$V = - \frac{\int_0^a a^3}{6\epsilon_0} \left[-\frac{1}{r'} \right]_{\infty}^r = \frac{\int_0^a a^3}{6\epsilon_0 r}$$

c) Gaussian surface of radius $r < a$.

$$\text{charge enclosed} = \frac{4\pi}{a^3} \int_0^r \left[\frac{a^3 r'^3}{3} - \frac{r'^6}{6} \right]_0^r$$

$$= \frac{4\pi}{a^3} \int_0^r \left[\frac{a^3 r'^3}{3} - \frac{r'^6}{6} \right]$$

$$= \frac{4\pi}{a^3} \times a^3 \int_0^r \left[\frac{2a^3 r'^3 - r'^6}{6} \right]$$

$$= \frac{2\pi}{3a^3} \int_0^r (2a^3 r'^3 - r'^6)$$

$$= \frac{2\pi r^3}{3a^3} \int_0^r (2a^3 - r'^3)$$

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$$\epsilon_0 E_r \cancel{4\pi r^2}^2 = \frac{2r^6 \pi \int_0 (2a^3 - r^3)}{3a^3}$$

$$E_r = \frac{r \int_0 (2a^3 - r^3)}{6a^3 \epsilon_0}$$

$$E = E_r \cdot 2r = \frac{r \int_0 (2a^3 - r^3)}{6a^3 \epsilon_0} 2r$$

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$V(r)$ inside the nucleus. $r < a$

$$V(r) = - \int E \cdot dl$$

$$= - \int_{\infty}^a E_{\text{ext}} \cdot dl - \int_a^r E_{\text{int}} \cdot dl$$

$$= - \int_{\infty}^a \frac{\int_0^a a^3}{6\epsilon_0 r^2} dr - \int_a^r \frac{r' \int_0 (2a^3 - r'^2)}{6a^3\epsilon_0} dr'$$

$$= + \frac{\int_0 a^3}{6\epsilon_0} \left[+\frac{1}{r} \right]_{\infty}^a - \frac{\int_0}{6a^3\epsilon_0} \int_a^r 2a^3 r' - r'^4 dr'$$
$$\frac{\int_0 a^2}{6\epsilon_0 r} - \frac{\int_0}{6a^3\epsilon_0} \left[\frac{a^3 r'^2}{2} - \frac{r'^5}{5} \right]_a^r$$

$$\frac{\int_0 a^2}{6\epsilon_0} - \frac{\int_0}{6a^3\epsilon_0} \left[\frac{a^3 r^2}{2} - \frac{r^5}{5} - a^5 + \frac{a^5}{5} \right]$$

$$\frac{\int_0}{6\epsilon_0} \left[a^2 - \frac{a^3 r^2}{a^3} + \frac{r^5}{5a^3} + \frac{a^5}{a^3} - \frac{a^5}{a^3 5} \right]$$

$$\frac{\int_0}{6\epsilon_0} \left[a^2 - r^2 + \frac{r^5}{5a^3} + a^2 - \frac{a^2}{5} \right]$$

$$V = \frac{\int_0}{6\epsilon_0} \left[\frac{9}{5} a^2 + \frac{r^5}{5a^3} - r^2 \right]$$

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$$Q_3: W = \frac{1}{2} [Q_1 V_1 + Q_2 V_2 + Q_3 V_3]$$

$$V_1 = \frac{k Q_2}{r_{12}} + \frac{k Q_3}{r_{13}}$$

$$r_{12} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$r_{13} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$Q_1 = 20 \text{ nC}$$

$$r_1 = (0, 0, 1)$$

$$Q_2 = 30 \text{ nC}$$

$$r_2 = (0, 1, 0)$$

$$Q_3 = -40 \text{ nC}$$

$$r_3 = (1, 0, 0)$$

$$V_1 = \frac{k}{\sqrt{2}} (30 \text{ nC} - 40 \text{ nC}) = -10 \text{ nC} \times \frac{k}{\sqrt{2}}$$

$$V_2 = \frac{k Q_1}{r_{12}} + \frac{k Q_3}{r_{23}}$$

$$= \frac{k}{\sqrt{2}} (20 \text{ nC} - 40 \text{ nC}) = -20 \text{ nC} \times \frac{k}{\sqrt{2}}$$

$$V_3 = \frac{k Q_2}{r_{23}} + \frac{k Q_1}{r_{13}}$$

$$= \frac{k}{\sqrt{2}} (30 \text{ nC} + 20 \text{ nC}) = 50 \text{ nC} \times \frac{k}{\sqrt{2}}$$

$$W = \frac{1}{2} [20 \text{ nC} \times -10 \text{ nC} + 30 \text{ nC} \times -20 \text{ nC} - 40 \text{ nC} \times 50 \text{ nC}] \frac{k}{\sqrt{2}}$$

$$= \frac{k}{2\sqrt{2}} [-200 \text{ nC}^2 - 600 \text{ nC}^2 - 2000 \text{ nC}^2]$$

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$$= \frac{k}{2\sqrt{2}} \times -2800 \times 10^{-18} =$$

$$= \frac{-2800 \times 10^{-18}}{2\sqrt{2} \times 4\pi \times 10^{-12} \times \epsilon_0} = -8.897 \times 10^{-6} \text{ J}$$

~~minus sign as work done against the electric field by external force~~

Negative sign indicates that work is done by the field.

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$$Q4: D = f^2 z \, \partial f + f \sin^2(\phi) \, \partial z$$

$$a) \oint_S D \cdot ds. \quad ds =$$

$$1 < f < 5, \quad 0 < z < 1 \quad 0 < \phi < 2\pi$$

Hollow cylinder top, bottom, inner surface, outer surface

~~top and bottom cancel as~~

$$\text{top} \int_{f=1}^5 \int_{\phi=0}^{2\pi} (f^2 z \, \partial f + f \sin^2(\phi) \, \partial z) \, dz \, dy \, dz$$

$$\text{bottom} \int_{f=1}^5 \int_{\phi=0}^{2\pi} (f^2 z \, \partial f + f \sin^2(\phi) \, \partial z) \, (-dz \, dy \, dz)$$

top + bottom = 0 thus cancelled.

inner. @ $f=1$.

$$\int_{\phi=0}^{2\pi} \int_{z=0}^1 (f^2 z \, \partial f + f \sin^2(\phi) \, \partial z) \, dz \, d\phi \, \partial f \quad -dz \, d\phi \, f \, \partial f$$

$$= \int_{\phi=0}^{2\pi} \int_{z=0}^1 f^2 z \cdot f \, d\phi \, dz.$$

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$$= - \int_0^3 \int_0^{2\pi} d\phi \int_{z=0}^1 z dz.$$

$$= -1 \cdot (2\pi) \times \frac{z^2}{2} \Big|_0^1 = -\pi.$$

outer (a) $\int = 5$.

$$\int_0^{2\pi} \int_{z=0}^1 (f^2 z \frac{\partial f}{\partial z}) + (f \sin^2(\phi) \frac{\partial z}{\partial z}) (dz d\phi \int dJ)$$

$$\int_0^3 \int_0^{2\pi} d\phi \int_{z=0}^1 z dz.$$

$$(5)^3 \times 2\pi \times \frac{z^2}{2} \Big|_0^1 = 125 \times \pi$$

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = 125\pi - \pi + 0 = 124\pi$$

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$$b) \int_V \nabla \cdot \mathbf{D} \, dv$$

$\nabla \cdot \mathbf{D}$ in cylindrical co-ordinates

$$\frac{1}{f} \frac{\partial (f D_r)}{\partial f} + \frac{1}{f} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$= \frac{1}{f} \frac{\partial (f^3 z)}{\partial f} + 0 + \frac{\partial (f \sin^2(\phi))}{\partial z}$$

$$= \frac{1}{f} (3f^2 z) + 0 = 3fz$$

$$\int_0^1 \int_0^{2\pi} \int_0^5 3fz \, d\phi \, dz \, df$$

$$3 \int_{f=1}^5 f^2 \, df \int_{z=0}^1 z \, dz \int_{\phi=0}^{2\pi} d\phi$$

$$\cancel{3} \times \frac{f^3}{\cancel{3}} \Big|_1^5 \times \frac{z^2}{\cancel{2}} \Big|_0^1 \times \cancel{2}\pi$$

$$\pi \times (125 - 1) \times 1 = 124\pi$$

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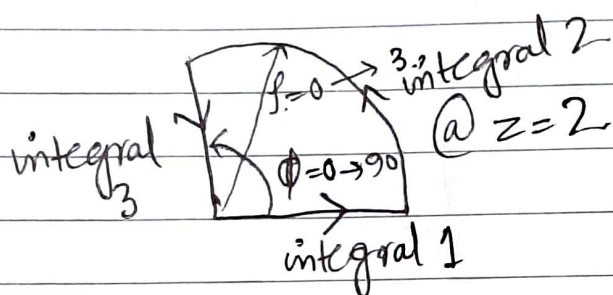
$$QS: A = f z^2 df + z \cos(\phi) d\phi + \int^2 \sin^2(\phi) dz$$

$$z = 2, 0 < f < 3, 0 < \phi < 90^\circ$$



Stokes Theorem

$$\oint E \cdot d\mathbf{l} = \int_S \nabla \times E \cdot d\mathbf{S}$$



integral 1 @ $\phi = 0$

$$\int_{f=0}^3 A \cdot df df = \int_0^3 f z^2 df = z^2 \times \frac{f^2}{2} \Big|_0^3$$

$$2^2 \times \left(\frac{9}{2} - 0 \right) = 18.$$

integral 3 @ $\phi = 90$.

$$\int_{f=3}^0 A \cdot df df = \int_3^0 f z^2 df = z^2 \times \frac{f^2}{2} \Big|_3^0$$

$$-18$$

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Integral 2: $\int_{\phi=0}^{\pi/2} A \cdot \oint d\phi, 2\phi = \int_{\phi=0}^{\pi/2} 2 \cos(\phi) \oint d\phi$

$\oint \int_0^{\pi/2} \cos(\phi) d\phi$

$2 \times 3 \left[\sin(\phi) \right]_0^{\pi/2} = 6 \times \left[\sin(\pi/2) - \sin 0 \right]$
 6×1

$\oint_{\ell} A \cdot d\ell = 18 - 18 + 6$
 $= 6$

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$$Q5: A = f z^2 \partial f + z \cos \phi + f^2 \sin(\phi) \partial z$$

$$z=0 \quad 0 < f < 3 \quad 0 < \phi < 90^\circ$$

$$\text{Stokes theorem: } \oint_L A \cdot d\mathbf{l} = \iint_S (\nabla \times A) \cdot d\mathbf{S}$$

$$\iint_S (\nabla \times A) \cdot d\mathbf{S}$$

Calculating $\nabla \times A$ in cylindrical co-ordinates.

$$\nabla \times A = \left[\frac{1}{f} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \partial f + \left[\frac{\partial A_f}{\partial z} - \frac{\partial A_z}{\partial f} \right] \partial \phi + \frac{1}{f} \left[\frac{\partial (f A_\phi)}{\partial f} - \frac{\partial A_f}{\partial \phi} \right] \partial z$$

$$d\mathbf{S} = f \, d\phi \, df \, \partial z$$

$$\iint_S (\nabla \times A) \cdot d\mathbf{S} = \iint_S \frac{1}{f} \left[\frac{\partial (f A_\phi)}{\partial f} - \frac{\partial A_f}{\partial \phi} \right] f \, d\phi \, df$$

$$\iint_S \left[\frac{\partial}{\partial f} (f z \cos \phi) - \frac{\partial}{\partial \phi} (f z^2) \right] d\phi \, df$$

$$\iint_S z \cos \phi - 0 \, d\phi \, df$$

$$= \int_{\phi=0}^{90} \cos \phi \, d\phi \int_{f=0}^3 df$$

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$$2 \times \sin \phi \Big|_0^{\pi/2} \times \int_0^3$$

$$2 \times (1 - 0) \times (3 - 0) = 6.$$

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = 6$$

thus proved Stokes theorem